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Basic Networking Commands

Aim: Study of various network commands used in Linux & Windows.

Linux Commands:

1) arp -a : ARP is short form of address resolution protocol, It will show the IP address of your computer along with IP address & MAC address of your router.

Output:

Internet Address	Physical address
172.16.11.109	00:27:0e:13:f7:8c
172.16.8.149	00:27:0e:13:f8:fe
172.16.9.212	7c:57:58:38:da:a3
172.16.8.186	7c:57:58:38:c6:24
172.16.8.249	00:27:0e:13:eb:3f

2) hostname : This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output : local host. local domain.

3) netstat : (network statistics) netstat display a variety of statistics about a computers active TCP/IP connection.

Output:

→ netstat

Active Connection

Packs	local address	Foreign address	state
TCP	172.16.10.31.7650	DESKTOP-IQPQVP8	Time-wait
TCP	172.16.10.31.5401	DESKTOP-IQPQVP8	Established

4. ipconfig /all : Gives detailed configuration information about TCP and IP connection.

Output:

Windows IP configuration

Host name .... : DESKTOP-IQPOVP8
Primary DNS suffix ... :
Node Type .... : No
IP Routing Enabled .. : No
WINS Proxy Enabled ... : No

Ethernet adapter Ethernet:

Description ..... : Realtek PCIe GbE Family Controller
Physical address ... : 00-27-0E-13-F2-F4
IPv4 Address .... : 172.16.9.55
Subnet Mask ... : 255.255.252.0

5. nslookup : Display DNS details of an specified host.

Output: → nslookup www.google.com
Server : unknown
Address : 172.16.8.1
Non-authoritative answer :
Name : www.google.com
Address : 2001:4860:4821:7700::
142.251.152.119
142.251.157.119

6. Pathping : It is a combination of ping and tracert command.

Output:
→ pathping www.google.com
Tracing route to google.com [142.16.8.1]
0 DESKTOP-V8I2B20 [142.16.8.1.52]
1 142.16.8.1
2 mail.tutorials.com [182.156.9.89]

8) ping : It is used to test the connectivity between a host.

Output :

→ ping google.com

pinging google.com [216.198.79.1] with 32 bytes of data.

Reply from 216.198.79.1 bytes = 32 time = 2ms TTL = 250

9) route PRINT

prints the routing table of the network.

Output :

→ route PRINT

IPV4 Route Table

Network destination	Network	Gateway	Interface
0.0.0.0	0.0.0.0	12.97.0.1	172.6.02.51

Result :

This is the basic networking commands been practiced.